Deep Latent Position Block Model (Deep LPBM)

Application to Brain Connectivity Graphs (HCP-100)

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1 Introduction

Graph data provide a compact way to describe complex systems where interactions matter as much as the individual elements. Classical probabilistic models such as the Stochastic Block Model (SBM) and its mixed-membership extension [1] capture community structure but rely on discrete assignments that restrict the expressiveness of the latent representation. In contrast, deep generative approaches such as Variational Graph Autoencoders [2] learn continuous embeddings but offer limited interpretability and no explicit modelling of block-level connectivity. The Deep Latent Position Block Model (Deep LPBM) [3] was introduced to reconcile these two perspectives by combining a block-structured decoder with a variational latent position model encoded through a graph convolutional network.

The goal of this project is to study, reproduce and critically evaluate Deep LPBM, both on controlled synthetic graphs and on functional connectivity networks from the HCP-100 dataset [4, 5, 6]. Beyond replicating the model's variational formulation, we examine the impact of practical choices that are often implicit in the original article, including initialization, negative sampling, and Monte-Carlo estimation of the ELBO. We also investigate the model's ability to recover the true number of clusters using information criteria. Finally, we apply Deep LPBM to human functional brain networks and assess whether the latent organisation learned by the model aligns with patterns of modularity and community structure consistently reported in the neuroscience literature [7, 8].

Contributions statement Lahna implemented the model and conducted the synthetic data experiments, as well as the preprocessing of the HCP-100 dataset. Adèle developed the negative sampling strategy and carried out the hyperparameter sensitivity analysis. Alessandro focused on the visualisation pipeline and performed additional experiments on both synthetic and HCP-100 data.

2 Deep LPBM: probabilistic model and variational inference

2.1 Generative model and notations

We consider an undirected binary graph $G = (V, E)$ with adjacency matrix $A \in \{0,1\}^{N\times N}$, where $A_{ij} = 1$ denotes an edge between i and j. The matrix is symmetric and satisfies $A_{ii} = 0$. Following the latent block modelling tradition, we assume that connectivity patterns arise from unobserved groups encoded through continuous latent variables. The graphical structure for each pair (i, j) is shown in Fig. A.1. Table 1 summarises the main notations.

Symbol	Description	Dimension
A	Adjacency matrix	$N \times N$
A_{ij}	Edge indicator	scalar
Q	Number of clusters	scalar
z_i	Latent Gaussian variable	$\mathbb{R}^{Q-1}$
η_i	Soft memberships	Δ^{Q-1}
Π	Block probability matrix	$Q \times Q$

Figure 1: Main notations in the Deep LPBM model.

Each node i is assigned a latent Gaussian vector z_i, then mapped to the probability simplex to obtain soft memberships η_i. This allows nodes to display mixed affiliations rather than hard assignments.

$$z_i \sim \mathcal{N}(0, I_{Q-1}), \qquad \eta_i = \mathrm{softmax}(z_i). \tag{1}$$

Cluster interactions are governed by a symmetric block matrix $\Pi \in [0,1]^{Q\times Q}$, where Π_{qr} encodes the probability of connection between groups q and r. Diagonal dominance yields assortative structure, while strong off-diagonal terms correspond to cross-cluster interactions.

Edges are then drawn independently according to

$$A_{ij} \mid \eta_i, \eta_j, \Pi \sim \mathrm{Bernoulli}(\eta_i^\top \Pi \eta_j), \qquad i < j, \tag{2}$$

which induces the full likelihood

$$p(A \mid \eta, \Pi) = \prod_{i<j} (\eta_i^\top \Pi \eta_j)^{A_{ij}} \left(1 - \eta_i^\top \Pi \eta_j\right)^{1-A_{ij}}. \tag{3}$$

The number of clusters Q is unknown and will be selected later through information criteria, but we treat it as fixed in what follows.

2.2 Intractable likelihood and the need for variational inference

A natural strategy to infer clustering would be to maximise the marginal log-likelihood

$$\log p(A \mid \Pi) = \log \int p(A, Z \mid \Pi)\, dZ, \tag{4}$$

but this quickly becomes out of reach: the softmax transformation ties all latent variables together, and the Bernoulli terms $\eta_i^\top \Pi \eta_j$ make the integral analytically hopeless. Trying to fall back on a classical EM algorithm does not help either. The posterior $p(Z \mid A, \Pi)$ does not separate over nodes, since each z_i influences every edge it participates in and is therefore entangled with all other z_j. Even a single-node update would require normalising a likelihood depending on the entire latent configuration. This global coupling between variables rules out closed-form optimisation and calls for a variational approximation instead.

2.3 Variational EM algorithm

We introduce an approximating distribution $R(Z)$ to surrogate the true posterior. For any choice of R, the marginal likelihood decomposes as:

$$\log p(A \mid \Pi) = \mathcal{L}(\Pi; R) + \mathrm{KL}(R(Z) \,\|\, p(Z \mid A, \Pi)), \tag{5}$$

where $\mathcal{L}(\Pi; R)$ is the Evidence Lower Bound (ELBO). The KL term is non-negative, so maximising the ELBO tightens the approximation of the true posterior. The ELBO is defined as

$$\mathcal{L}(\Pi; R) = \mathbb{E}_{R(Z)}\left[\log \frac{p(A, Z \mid \Pi)}{R(Z)}\right]. \tag{6}$$

2.4 Variational family and resulting ELBO

To keep the inference tractable, we adopt a mean-field approximation and choose Gaussian node-wise variational factors $R_{\phi,i}(z_i) = \mathcal{N}(z_i; \mu_{\phi,i}, \sigma^2_{\phi,i} I_d)$:

$$R_\phi(Z) = \prod_{i=1}^{N} R_{\phi,i}(z_i), \tag{7}$$

The encoder network outputs $(\mu_{\phi,i}, \sigma_{\phi,i})$ for each node, enabling amortised inference. Sampling uses the reparameterisation trick, and the KL divergence admits a closed form. Under these assumptions, the ELBO becomes (proof in A.3):

$$\begin{aligned}\mathcal{L}(\Pi; R_\phi) = &\sum_{j<i} \mathbb{E}_{R_\phi(Z)}[\log p(A_{ij} \mid \eta_i, \eta_j, \Pi)] \\ &- \sum_{i=1}^{N} \mathrm{KL}(R_{\phi,i}(z_i) \,\|\, p(z_i)).\end{aligned} \tag{8}$$

This objective balances edge reconstruction and regularisation of the latent variables, and forms the basis of the training procedure.

2.5 Links with other models

Deep LPBM includes several well-known models as special cases. If memberships collapse to one-hot vectors, the model reduces to the classical SBM. If $\Pi = I$, the decoder becomes a dot-product model, equivalent to the VGAE decoder. If Π is fully unconstrained, the model aligns with mixed-membership SBM and with the latent position formulation of GRDPG. Deep LPBM therefore bridges discrete community detection, soft memberships, and continuous latent geometry within a single probabilistic framework.

3 Practical implementation and training scheme

3.1 K-means initialization

Because the optimisation landscape of Deep LPBM is highly sensitive to its starting point, we follow the recommendation of the original article and rely on an initial K-means step. We cluster the rows of the adjacency matrix into Q groups, obtaining a coarse partition that we immediately soften into membership vectors η_0 through a small uniform smoothing to avoid zero entries and keep all nodes strictly inside the simplex. The corresponding latent logits are then constructed by simple log-ratios,

$$z_{0,iq} = \log(\eta_{0,iq} + \tau) - \log(\eta_{0,iQ} + \tau), \tag{9}$$

where τ is a stabilising constant. An initial block matrix Π_0 is finally estimated in an SBM-like manner by weighting edges with $\eta_0\eta_0^\top$ and symmetrising. This procedure stays close to the spirit of the original model while improving numerical stability, and it consistently leads to faster and more reliable convergence.

3.2 Encoder and decoder architecture

Encoder. The encoder is a two-layer graph convolutional network operating on the symmetrized adjacency matrix. The GCN first ensures the normalization of the adjacency matrix:

$$\tilde{A} = A + I_N, \qquad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}, \qquad \tilde{L} = \tilde{D}^{-1/2} \tilde{A}\, \tilde{D}^{-1/2}.$$

Given the normalized operator $\tilde{L}$, the encoder computes

$$H_1 = \mathrm{ReLU}(\tilde{L} W_0), \quad \mu_\phi = \tilde{L} H_1 W_\mu, \quad \log \sigma_\phi = \tilde{L} H_1 W_\sigma, \tag{10}$$

where W_0, W_μ and W_σ are learned parameter matrices. This parametrization provides node-wise Gaussian variational factors $R_{\phi,i}(z_i) = \mathcal{N}(\mu_{\phi,i}, \sigma^2_{\phi,i} I)$ and amortizes inference over the graph structure. The latent representations z_i are then sampled by the reparameterization trick, and the soft memberships $\eta_i = \mathrm{softmax}(z_i)$ transport these embeddings to the simplex.

Decoder. Edges are reconstructed through an interaction matrix Π acting on the soft memberships:

$$p(A_{ij} = 1 \mid \eta_i, \eta_j, \Pi) = \eta_i^\top \Pi \eta_j. \tag{11}$$

Since Π must lie in $(0,1)^{Q\times Q}$ but it is inconvenient to optimize under box constraints, we instead learn an unconstrained matrix $\tilde{\Pi}$ and map it to the probability space using the smooth bijection :

$$\Pi = f(\tilde{\Pi}), \qquad f(x) = \tfrac{1}{2} + \tfrac{1}{\pi} \arctan(x). \tag{12}$$

This transformation avoids saturation at boundary values, and ensures that the decoder naturally interpolates between assortative and disassortative structures.

3.3 Practical training objective: Monte Carlo ELBO and negative sampling

In practice, Deep LPBM is trained using a Monte Carlo estimate of the ELBO. The KL term is straightforward thanks to the Gaussian variational family and takes the form:

$$\mathrm{KL}(R_{\phi,i} \,\|\, p(z_i)) = \frac{1}{2} \sum_{d=1}^{Q-1} \left(\mu_{id}^2 + \sigma_{id}^2 - \log \sigma_{id}^2 - 1\right), \quad (13)$$

But the reconstruction term $\mathbb{E}_{R_\phi(Z)}[\log p(A_{ij} \mid \eta_i, \eta_j, \Pi)]$ cannot be computed in closed form because $\eta_i = \mathrm{softmax}(z_i)$ is a nonlinear function of a Gaussian variable. We therefore draw one reparameterised sample per node, and set $\eta_i^{(s)} = \mathrm{softmax}(z_i^{(s)})$, from which the reconstruction term is estimated empirically.

$$z_i^{(s)} = \mu_i + \sigma_i \odot \epsilon_i^{(s)}, \qquad \epsilon_i^{(s)} \sim \mathcal{N}(0, I), \quad (14)$$

Evaluating this term exactly would require looping over all $\binom{N}{2}$ node pairs, which is unnecessary and prohibitive on sparse graphs. As in many graph autoencoders, we adopt negative sampling. We denote by $E^+ = \{(i,j) : A_{ij} = 1\}$ the set of observed edges and by E^- a small random sample of non-edges (typically a few negatives per positive). The reconstruction term is then approximated by 15 which provides an unbiased stochastic estimate while reducing complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(|E^+|)$.

$$\widehat{\mathcal{L}}_{\mathrm{rec}} = \sum_{(i,j)\in E^+} \log(\eta_i^\top \Pi \eta_j) + \sum_{(k,\ell)\in E^-} \log\bigl(1 - \eta_k^\top \Pi \eta_\ell\bigr), \quad (15)$$

This final Monte Carlo ELBO is optimised jointly over the encoder parameters ϕ and the unconstrained block matrix $\tilde{\Pi}$ using Adam, yielding a scalable amortised variational EM procedure and is given by:

$$\widehat{\mathcal{L}}(\Pi; R_\phi) = \widehat{\mathcal{L}}_{\mathrm{rec}} - \sum_{i=1}^{N} \mathrm{KL}(R_{\phi,i} \,\|\, p(z_i)). \quad (16)$$

3.4 Model selection and training procedure

Selecting the number of clusters Q is performed using information criteria computed after training. For each candidate Q, we record the final ELBO value $\widehat{\mathcal{L}}_Q$ and apply AIC, BIC, and ICL,

$$\begin{aligned} \mathrm{AIC}(Q) &= -2\,\widehat{\mathcal{L}}_Q + 2\,k_Q, \\ \mathrm{BIC}(Q) &= -2\,\widehat{\mathcal{L}}_Q + k_Q \log N, \\ \mathrm{ICL}(Q) &= \mathrm{BIC}(Q) - 2\,\mathbb{E}_{R_\phi(Z)}[H(\eta)], \end{aligned} \quad (17)$$

where $H(\eta_i) = -\sum_{q=1}^{Q} \eta_{iq} \log \eta_{iq}$ is the membership entropy. These scores highlight different preferences: AIC often favours richer models, BIC favors simpler ones, and ICL penalizes diffuse cluster assignments.

Post-processing of small clusters (extension). In practice, AIC may overestimate Q by promoting partitions containing very small clusters. To avoid such artefacts, we apply a lightweight post-processing step: clusters with size below a minimal threshold are collapsed by reassigning their nodes to the next most probable cluster. This adjustment preserves the interpretability of the partition and stabilizes the computation of the information criteria, especially in high-Q regimes.

Training setup. For each tested value of Q, the model is trained by alternating updates of the encoder parameters ϕ and the unconstrained interaction matrix $\tilde{\Pi}$ using Adam. Negative sampling is used throughout to evaluate the reconstruction term, and reparameterized Gaussian sampling is used to propagate gradients through the latent variables. The value of Q retained for subsequent analysis is the one minimizing the selected information criterion. The full training workflow, including initialization, variational updates, sampling strategy, and model selection, is summarized in Algorithm 1 in A.2.

4 Datasets and preprocessing

4.1 Synthetic data generation

We evaluate Deep LPBM on synthetic graphs generated from controlled stochastic block structures (assortative, disassortative, or hub-like; see Appendix A.4). For each dataset, we fix the number of nodes N, the true number of clusters Q, and a symmetric block matrix $\Pi^\star$ built from two probabilities: a high value β defining the dominant connections of the chosen regime, and a low baseline ε elsewhere. The matrix is then symmetrised.

Node memberships follow the partially mixed structure of the original paper. Each node is assigned a true cluster y_i, and its soft membership vector is defined by $\eta_i^\star = \zeta\, e_{y_i} + (1-\zeta)\frac{1}{Q}\mathbf{1}$, so that ζ controls the sharpness of the communities, from perfectly separated ($\zeta = 1$) to increasingly overlapping. Edge probabilities are computed as $P = \eta^\star \Pi^\star \eta^{\star\top}$, and the adjacency matrix A is sampled nodewise from Bernoulli draws. This setup yields both ground-truth partitions and soft memberships, allowing us to evaluate models via ARI for hard labels and PMI for soft assignments. By varying Q, N, ζ, and the underlying structural regime, we obtain a range of scenarios from clean block models to more challenging mixed settings, providing a controlled testbed for assessing Deep LPBM.

4.2 Real data: HCP-100 functional brain networks (Glasser–180)

We evaluate Deep LPBM on functional connectivity graphs derived from the Human Connectome Project (HCP) 100 unrelated subjects. The dataset, released by Sreedharan and Ravindra [4], is constructed from resting-state fMRI signals of the HCP 100-subject cohort [5] and parcellated using the Glasser atlas [6].There are 2 sessions per subject and 2 scans per subject, and thus in total 400 data point with $N = 180$ cortical regions.

Starting from voxel-wise fMRI time series, the authors apply standard preprocessing steps (including cortical extraction, global signal regression, and parcel averaging), then compute correlation matrices between parcels. These matrices are converted to binary adjacency graphs using spanning-tree plus k-nearest neighbour augmentation. The dataset provides these graphs in edge-list format; we reconstruct the adjacency matrix, symmetrize it, and enforce $A_{ii} = 0$. Since no node features are supplied, identity features $X = I_N$ are used throughout.

4.3 Baselines and evaluation metrics

To contextualize the performance of Deep LPBM, we compare it with standard community-detection methods implemented in `graspologic` and `scikit-learn`, ensuring that differences arise from the modeling rather than the software.

Baselines. Spectral Clustering serves as a classical reference for assortative structure, and its soft variant provides a continuous embedding close in spirit to our latent-space formulation. Louvain and Leiden offer strong empirical baselines through modularity optimisation, although without probabilistic grounding. The Stochastic Block Model (SBM) is a natural point of comparison, as Deep LPBM reduces to an SBM when memberships become one-hot, making this baseline particularly relevant on synthetic block-structured graphs.

Synthetic-data metrics. Since ground truth is available, clustering quality is evaluated using the Adjusted Rand Index (ARI), which compares pairwise co-clustering decisions and varies between 0 (random) and 1 (perfect match). We also report the Normalized Mutual Information (NMI), though it is not central to our conclusions. For soft assignments, we use the Partial Membership Index (PMI), which compares co-clustering probabilities implied by η and $\hat{\eta}$:

$$PMI = \sqrt{\frac{2}{N(N-1)} \sum_{i \le j} \left| (\eta\eta^\top - \hat{\eta}\hat{\eta}^\top)_{ij} \right|}, \qquad (18)$$

with $PMI \in [0, 1]$ and 0 indicating identical soft partitions. PMI is more sensitive than ARI to partial overlaps and is well aligned with the continuous nature of Deep LPBM.

Real-data metrics. For fMRI-derived networks, no ground truth is available, so we rely on associative and disassociative connectivity scores, which quantify how the learned partition aligns with observed connectivity. For a block matrix Π, they are defined as

$$ACS = \frac{1}{Q} \sum_{q=1}^{Q} \Pi_{qq}, \qquad DCS = \frac{2}{Q(Q-1)} \sum_{q<r} \Pi_{qr}. \qquad (19)$$

High ACS reflects assortative structure, while high DCS suggests cross-cluster interactions. These metrics do not measure accuracy but provide an interpretable summary of the structural regime captured by the model—particularly relevant in brain networks, where communities arise from distributed functional organisation rather than discrete labels.

5 Experimental Results

5.1 Synthetic graphs: comparison with baseline methods

We evaluate Deep LPBM on synthetic assortative graphs with $N = 250$, $Q = 5$, and $(\beta, \varepsilon) = (0.6, 0.05)$, while varying the mixing parameter $\zeta \in [0.6, 1]$ to progressively blur the community structure. The baselines described in Section 4.3 are included for comparison, and examples of soft and hard assignments are shown in Figures 2 and A.2.

Clustering accuracy (ARI). Figure A.3 (top) shows that all methods behave similarly when $\zeta \approx 1$, recovering the block structure almost perfectly. As ζ decreases and communities overlap, the gap between methods widens. Classical approaches (SBM, spectral clustering) remain robust down to moderately mixed regimes, whereas Deep LPBM deteriorates more rapidly, dropping to an ARI near 0.55 around $\zeta = 0.65$. This suggests that the geometric component of the model is more sensitive to blurred boundaries than strictly discrete models.

Soft membership estimation (PMI). The PMI results in Figure A.3 (bottom) give a more nuanced picture. Deep LPBM maintains a roughly constant error across all values of ζ, indicating that it captures a stable underlying geometry even when hard clustering becomes ambiguous. Methods relying on spectral embeddings perform well only when clusters are distinct; as overlap increases, they tend to oversmooth and lose the fine-grained membership structure. Soft spectral clustering is the only baseline whose behaviour remains comparable to Deep LPBM in this metric.

Overall, the figures illustrate a clear trade-off: Deep LPBM is less competitive for recovering sharp partitions and number of clusters but is more consistent when evaluating continuous membership structure, which aligns with its design as a hybrid geometric–block model rather than a pure clustering algorithm.

5.2 Sensitivity to hyperparameters

We analyse the effect of several key hyperparameters on synthetic graphs, using PME as our main metric (see Section 4.3). Results are averaged over multiple initialisations.

Number of hidden units. We compare encoder widths of 16, 32 and 64 units (Table 1). A larger layer increases expressiveness but can destabilise training. In practice, 32 units offer the best compromise: performance matches or slightly improves over 16 units, while avoiding the variability observed with 64. Since runtimes are similar, we adopt 32 units as the default.

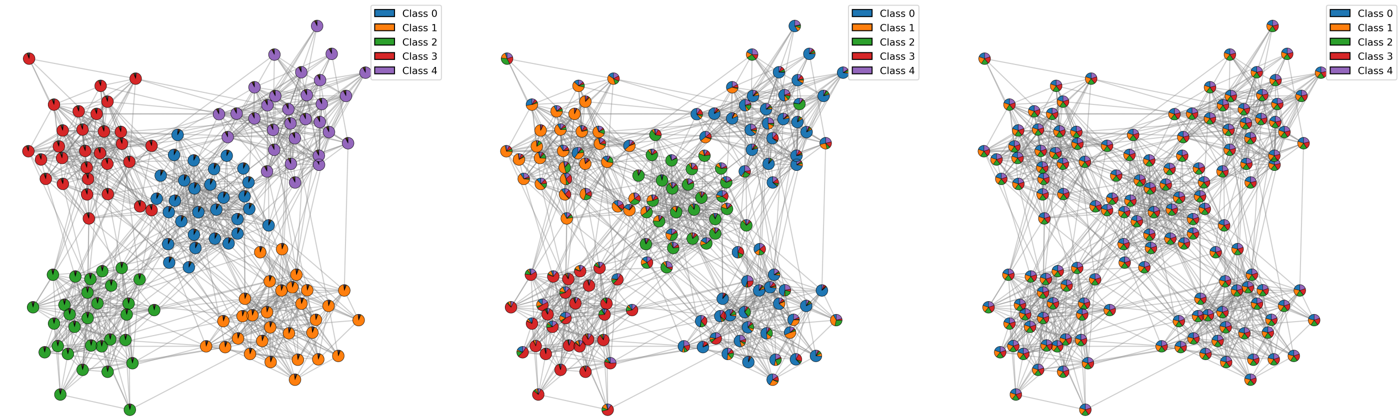


Figure 2: Comparison of soft community structure on a synthetic graph ($N = 150$, $\beta = 0.6$, $Q = 5$). Left: ground-truth partial memberships. Center: soft assignments recovered by DeepLPBM. Right: soft spectral clustering baseline.

	16 units	32 units	64 units
PME (H)	0.35 ± 0.0065	0.35 ± 0.0066	0.35 ± 0.0143
Runtime (s)	18.3 ± 0.84	19.5 ± 0.70	19.0 ± 0.74

Table 1: Effect of encoder hidden dimension on PME score and runtime.

	$N = 250$	$N = 700$	$N = 1200$
PME (H)	0.367 ± 0.035	0.284 ± 0.050	0.256 ± 0.056
Runtime (s)	18.8 ± 4.22	23.5 ± 0.52	38.1 ± 0.94

Table 3: Effect of the number of nodes N on PME score and runtime.

Learning rate. We evaluate learning rates 10^{-3}, 5×10^{-3} and 10^{-2} (Table 2). The smallest value converges slowly and unreliably, whereas the two larger ones yield stable behaviour and similar PME scores. We therefore retain 5×10^{-3} as a balanced setting.

	10^{-3}	5×10^{-3}	10^{-2}
PME (H)	0.38 ± 0.0381	0.35 ± 0.0068	0.35 ± 0.0066
Runtime (s)	31.0 ± 10.43	18.7 ± 0.52	18.1 ± 0.63

Table 2: Effect of learning rate on PME score and runtime.

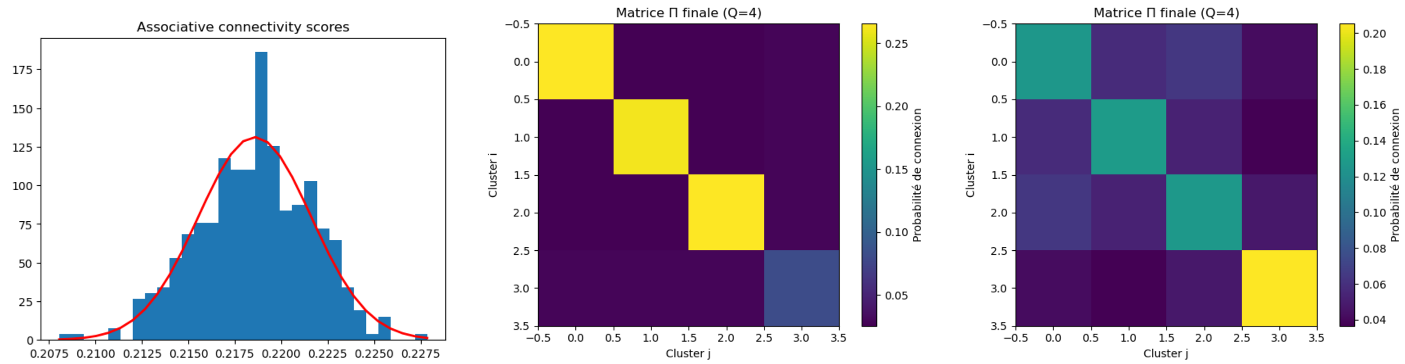
Model selection for Q. We finally test AIC, BIC and ICL for recovering the true number of clusters. AIC is by far the most reliable, selecting the correct Q in 93% of runs, whereas BIC almost never succeeds and ICL only in about 44% of cases. This mirrors the behaviour reported in the original Deep LPBM article, where AIC also emerges as the most appropriate criterion. Accordingly, we use AIC in all subsequent experiments.

Impact of the number of nodes N. Increasing the graph size consistently improves the quality of the inferred memberships, with the partial membership error decreasing as N grows (Table 3). Larger graphs provide more structural information, allowing Deep LPBM to better identify the underlying block structure. However, this gain in accuracy comes at the cost of higher computation time, which increases markedly for large N due to the heavier GCN message passing and reconstruction steps. Overall, Deep LPBM benefits statistically from larger graphs but exhibits a clear scalability–accuracy trade-off.

5.3 Brain connectivity graphs

We apply Deep LPBM to the 400 functional connectivity graphs of the HCP-100 dataset, each with $N = 180$ cortical regions. Each scan is fitted independently using the training scheme of Section 3. In all cases, the model selects the same number of clusters, $Q^\star = 4$, suggesting that at the Glasser–180 resolution the functional connectome can be summarised by a coarse four-module structure.

To characterise the corresponding block structure, we compute the associative and disassociative scores of Eq. 19 on the learned Π across subjects. We obtain $ACS = 0.22 \pm 0.03$, $DCS = 0.027 \pm 0.0005$, with a single clear outlier. The narrow spread of these values shows that the fitted block matrices are highly reproducible across subjects, and that within-cluster connectivity remains systematically stronger than cross-cluster links. Figure 3 illustrates the distribution of associative scores, the average block matrix, and the outlier. To examine node-wise stability, we define for each region a switch score $\mathrm{Switch}(i) = 1 - \frac{n_{\max}(i)}{400}$, where $n_{\max}(i)$ is the number of times node i is assigned to its most frequent cluster over the 400 scans. By construction, $\mathrm{Switch}(i) = 0$ corresponds to a perfectly stable assignment (same cluster for all subjects), while random labelling across $Q = 4$ clusters would give an expected value of $1 - \frac{1}{Q} = 0.75$. In our results, most regions fall between 0.45 and 0.65, indicating neither rigid nor random behaviour but a moderate, structured variability. Visual inspection of the soft assignments (Figure A.4) shows that high-switch regions tend to lie at the boundaries between clusters, where communities overlap.



(a) Distribution of associative scores across subjects.

(b) Population-averaged block matrix.

(c) Block matrix for the single outlier.

Figure 3: Connectivity structure recovered from HCP-100 functional networks. DeepLPBM systematically identifies four clusters and a strongly assortative organization.

This pattern is consistent with the hierarchical and partially overlapping organisation reported in the literature. Large-scale modular structure is a robust feature of resting-state functional networks [7], but nodal community memberships are known to be less clear-cut in integrative or hub regions [8]. In our case, three clusters appear relatively cohesive, while the fourth is more diffuse and intertwined with the others. This can be interpreted as one level in a multi-scale hierarchy: depending on thresholding or representation, these mixed zones could either form a separate integrative module or be further subdivided into smaller subclusters. The behaviour of Deep LPBM on HCP-100 is therefore coherent with existing views of brain modularity as a graded and multi-level phenomenon, rather than a single fixed partition.

6 Discussion

The synthetic experiments offer a mixed but consistent picture. When communities are well separated, Deep LPBM recovers the true structure with high accuracy, as in [3]. As memberships begin to overlap, performance drops faster than for discrete models: ARI declines while partial membership error plateaus. The encoder tends to oversmooth representations, whereas the block decoder favours sharp boundaries; when these forces conflict, optimisation often settles on a coarse latent geometry rather than refined soft assignments.

Cluster-number selection reveals a similar tension. Although the original article reports stable recovery of Q, our results show a trade-off: aggressive initialisation can improve identification of Q but harm clustering quality, while looser settings give better ARI but occasionally misestimate model order. BIC and ICL remain overly conservative; AIC behaves best but still depends on optimisation stability. Our extensions—collapsing tiny clusters and adding negative sampling—substantially improved robustness, especially under moderate noise.

Results on real data stand in contrast to these limitations. On HCP-100, Deep LPBM consistently converges to the same four-cluster organisation across all scans, with strongly assortative block matrices and very low inter-subject variability. This aligns with resting-state findings: large-scale cortical modules are reproducible [7], while connector regions show more diffuse affiliations [8], reflected here in the less sharply defined fourth cluster.

Overall, Deep LPBM works best when modular organisation is present but not overly noisy. In such cases, its hybrid structure—continuous geometry with interpretable blocks—is effective. Under heavy overlap, however, the same hybrid design becomes limiting, though our modifications alleviate part of this behaviour.

7 Conclusion

This re-implementation of Deep LPBM clarifies the model's strengths and limits. On synthetic graphs, it recovers block structure when communities are distinct and the number of nodes is sufficient, but performance deteriorates as memberships overlap and the latent geometry becomes harder to reconcile with block constraints. Model selection is more delicate than suggested in the original article, with AIC providing the most reliable behaviour.

On real functional connectivity networks, the model behaves differently: it consistently extracts a four-cluster structure aligned with known large-scale modular patterns in the human connectome. This contrast suggests that Deep LPBM is well suited to datasets with modular but not strictly discrete organisation, where a continuous latent space offers interpretive value.

Potential extensions include weighted decoders for non-binary networks, additional entropy or sparsity regularisation to stabilise memberships, and multi-subject or hierarchical variants tailored to neuroimaging. Such developments may improve robustness while preserving the model's main advantage: linking interpretable block structure with flexible latent geometry.

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A Appendix

A.1 Graphical Model

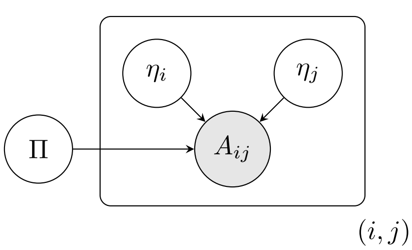


Figure A.1: Graphical representation of the Deep LPBM generative mechanism.

Algorithm 1: Training and model selection for Deep LPBM

Input: Adjacency matrix A, features X (or I_N), candidate values $\mathcal{Q}$, initialization epochs T_{init}, training epochs T, learning rate lr, negative sampling ratio ρ, MC samples S

Output: Best (ϕ, Π), latent variables (z_i, η_i) and chosen $Q^\star$

for $Q \in \mathcal{Q}$ **do**

`// K-means based initialization of latent positions`

Construct $Z^0 = f^{-1}(C^{\text{KMeans}})$ (inverse of the softmax mapping);

`// Initialisation of the GCN encoder`

Initialize encoder parameters ϕ and unconstrained block matrix $\tilde{\Pi} \in \mathbb{R}^{Q \times Q}$;

for *epoch* $= 1$ **to** T_{init} **do**

$(\mu, \sigma) \leftarrow \text{Encoder}(A, X; \phi)$;

$\ell_{\text{init}}(\phi) \leftarrow \frac{1}{N} \sum_{i=1}^{N} \left\|\mu_i - z_i^0\right\|_2^2 + \left\|\sigma_i^2 - 0.01\right\|_2^2$;

Update ϕ by stochastic gradient descent on $\ell_{\text{init}}(\phi)$;

`// Estimation of Deep LPBM`

for *epoch* $= 1$ **to** T **do**

$(\mu, \log \sigma^2) \leftarrow \text{Encoder}(A, X; \phi)$;

Sample $z_i^{(s)} = \mu_i + \sigma_i \odot \epsilon_i^{(s)}$, $\epsilon_i^{(s)} \sim \mathcal{N}(0, I)$, and set $\eta_i^{(s)} = \text{softmax}(z_i^{(s)})$ for $s = 1, \ldots, S$;

Map $\tilde{\Pi}$ to a valid block matrix $\Pi \leftarrow f(\tilde{\Pi})$ with $f(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x)$;

Extract positive edges $E^+ = \{(i, j) : A_{ij} = 1, \, i < j\}$ and sample a set of non-edges $E^- = \texttt{SampleNonEdges}(A, \rho)$;

Estimate the reconstruction term $\widehat{\mathcal{L}}_{\text{rec}}$ on $E^+ \cup E^-$ using the Monte Carlo samples $\eta^{(s)}$;

Compute the closed-form KL term $\mathcal{L}_{\text{KL}} = \frac{1}{2} \sum_{i,d} (\mu_{id}^2 + \sigma_{id}^2 - \log \sigma_{id}^2 - 1)$;

Practical ELBO: $\widehat{\mathcal{L}} = \widehat{\mathcal{L}}_{\text{rec}} - \mathcal{L}_{\text{KL}}$;

Update ϕ and $\tilde{\Pi}$ via Adam on the loss $-\widehat{\mathcal{L}}$;

`// Post-processing and information criteria (ours)`

Compute final memberships $\eta = \text{softmax}(\mu)$ and apply $\eta \leftarrow \texttt{CollapseSmallClusters}(\eta)$;

Compute AIC, BIC, ICL from $\widehat{\mathcal{L}}$, η and Π and store them for this Q;

Select $Q^\star$ according to the chosen criterion (AIC/BIC/ICL);

return (ϕ, Π) and (z_i, η_i) corresponding to $Q^\star$;

A.2 Training Algorithm

A.3 Derivation of the ELBO

By definition of the ELBO, we have

$$\mathcal{L}(\Pi; R_\phi) = \mathbb{E}_{R_\phi(Z)}\big[\log p(A, Z \mid \Pi) - \log R_\phi(Z)\big].$$

Using the factorization $p(A, Z \mid \Pi) = p(A \mid Z, \Pi)\, p(Z)$, this becomes

$$\mathcal{L}(\Pi; R_\phi) = \mathbb{E}_{R_\phi(Z)}\big[\log p(A \mid Z, \Pi) + \log p(Z) - \log R_\phi(Z)\big].$$

By linearity of the expectation,

$$\mathcal{L}(\Pi; R_\phi) = \mathbb{E}_{R_\phi(Z)}[\log p(A \mid Z, \Pi)] + \mathbb{E}_{R_\phi(Z)}[\log p(Z)] - \mathbb{E}_{R_\phi(Z)}[\log R_\phi(Z)].$$

(1) Decomposition of the reconstruction term. By conditional independence of the edges given Z and Π,

$$p(A \mid Z, \Pi) = \prod_{i<j} p(A_{ij} \mid \eta_i, \eta_j, \Pi).$$

Taking the logarithm gives

$$\log p(A \mid Z, \Pi) = \sum_{i<j} \log p(A_{ij} \mid \eta_i, \eta_j, \Pi),$$

and therefore

$$\mathbb{E}_{R_\phi(Z)}[\log p(A \mid Z, \Pi)] = \sum_{i<j} \mathbb{E}_{R_\phi(Z)}\big[\log p(A_{ij} \mid \eta_i, \eta_j, \Pi)\big].$$

(2) Decomposition of the KL term. Using the factorization of the prior and the variational distribution,

$$p(Z) = \prod_{i=1}^{N} p(z_i), \qquad R_\phi(Z) = \prod_{i=1}^{N} R_{\phi,i}(z_i),$$

we have

$$\log p(Z) = \sum_{i=1}^{N} \log p(z_i), \qquad \log R_\phi(Z) = \sum_{i=1}^{N} \log R_{\phi,i}(z_i).$$

Thus

$$\mathbb{E}_{R_\phi(Z)}[\log p(Z) - \log R_\phi(Z)] = \sum_{i=1}^{N} \mathbb{E}_{R_\phi(Z)}\big[\log p(z_i) - \log R_{\phi,i}(z_i)\big].$$

By factorization of $R_\phi(Z)$, the expectation with respect to Z reduces to an expectation with respect to z_i under $R_{\phi,i}$:

$$\mathbb{E}_{R_\phi(Z)}\big[\log p(z_i) - \log R_{\phi,i}(z_i)\big] = \mathbb{E}_{R_{\phi,i}(z_i)}\big[\log p(z_i) - \log R_{\phi,i}(z_i)\big].$$

Hence

$$\mathbb{E}_{R_\phi(Z)}[\log p(Z) - \log R_\phi(Z)] = \sum_{i=1}^{N} \mathbb{E}_{R_{\phi,i}(z_i)}\big[\log p(z_i) - \log R_{\phi,i}(z_i)\big].$$

Recognizing the negative Kullback–Leibler divergence,

$$\mathbb{E}_{R_{\phi,i}}\big[\log p(z_i) - \log R_{\phi,i}(z_i)\big] = -\mathrm{KL}\big(R_{\phi,i}(z_i) \,\|\, p(z_i)\big),$$

we obtain

$$\mathbb{E}_{R_\phi(Z)}[\log p(Z) - \log R_\phi(Z)] = -\sum_{i=1}^{N} \mathrm{KL}\big(R_{\phi,i}(z_i) \,\|\, p(z_i)\big).$$

(3) Final expression. Combining the reconstruction term and the KL term, we get

$$\mathcal{L}(\Pi; R_\phi) = \sum_{i<j} \mathbb{E}_{R_\phi(Z)}\big[\log p(A_{ij} \mid \eta_i, \eta_j, \Pi)\big] - \sum_{i=1}^{N} \mathrm{KL}\big(R_{\phi,i}(z_i) \,\|\, p(z_i)\big),$$

which is the desired decomposition.

A.4 Connectivity structures used in synthetic data

The three generative modes used in our synthetic experiments correspond to the following block connectivity matrices, parameterized by (β, ε) with $\beta > \varepsilon$.

Assortative structure. Clusters are densely connected internally and weakly connected across groups:

$$\Pi_{\text{assort}} = \begin{pmatrix} \beta & \varepsilon & \cdots & \varepsilon \\ \varepsilon & \beta & \cdots & \varepsilon \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon & \varepsilon & \cdots & \beta \end{pmatrix}.$$

Disassortative structure. Between-cluster links are favored over within-cluster interactions:

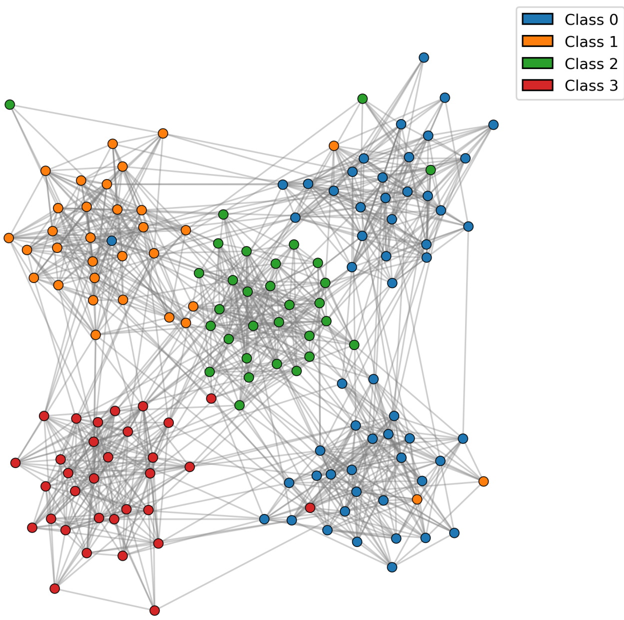
$$\Pi_{\text{disassort}} = \begin{pmatrix} \varepsilon & \beta & \cdots & \beta \\ \beta & \varepsilon & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \varepsilon \end{pmatrix}.$$

Hub structure. A designated cluster strongly connects to all others, while non-hub groups remain mostly assortative:

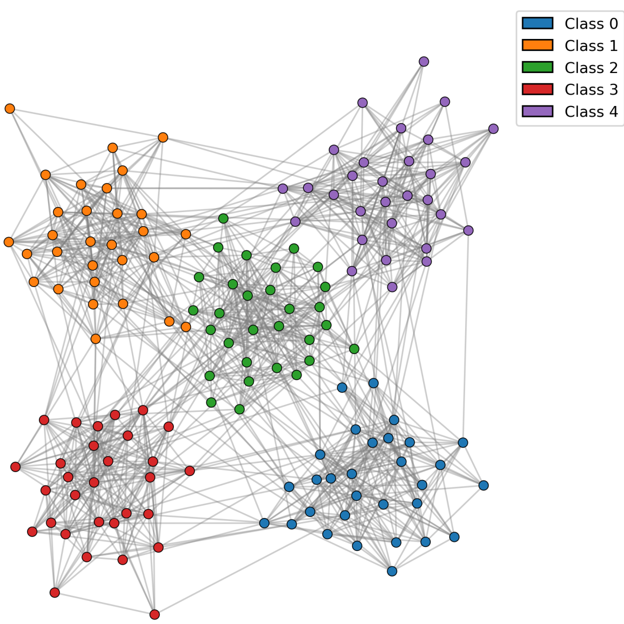
$$\Pi_{\text{hub}} = \begin{pmatrix} \beta & \beta & \beta & \cdots & \beta \\ \beta & \beta & \varepsilon & \cdots & \varepsilon \\ \beta & \varepsilon & \beta & \cdots & \varepsilon \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \beta & \varepsilon & \varepsilon & \cdots & \beta \end{pmatrix}.$$

These structures reproduce the synthetic settings introduced in the original Deep LPBM article and used in our experiments.

A.5 Clusters from synthetic Data



(a) Hard labels for DeepLPBM



(b) Hard labels for Soft Spectral Clustering

Figure A.2: For visualization purposes, $N = 150$, $\beta = 0.6$, $Q = 5$

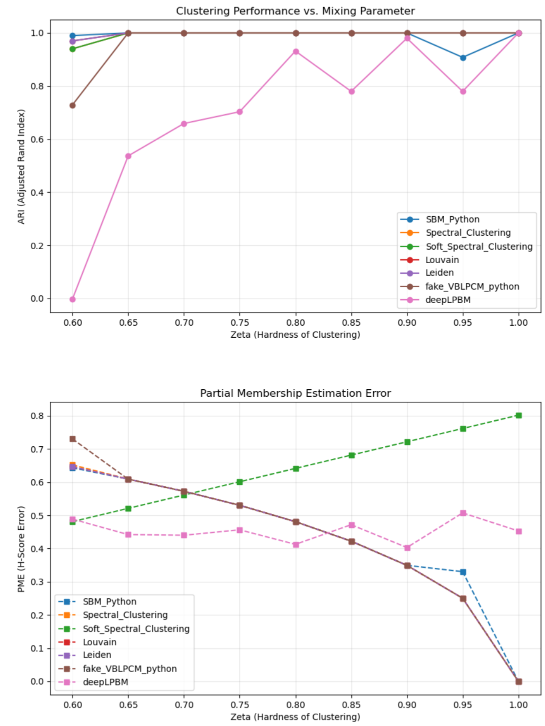
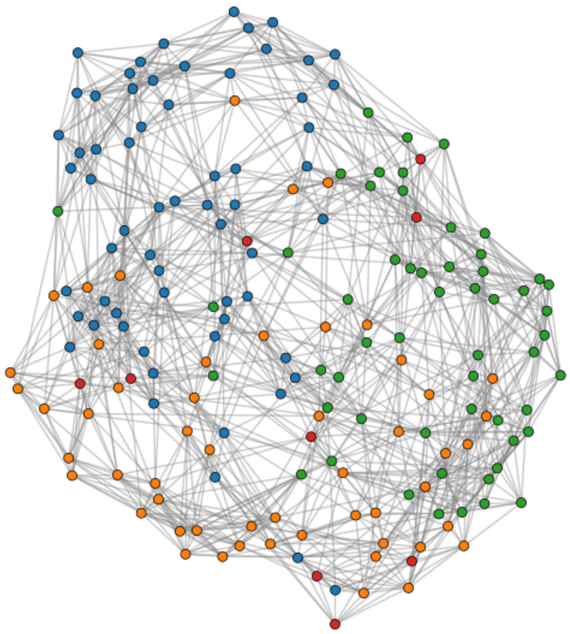
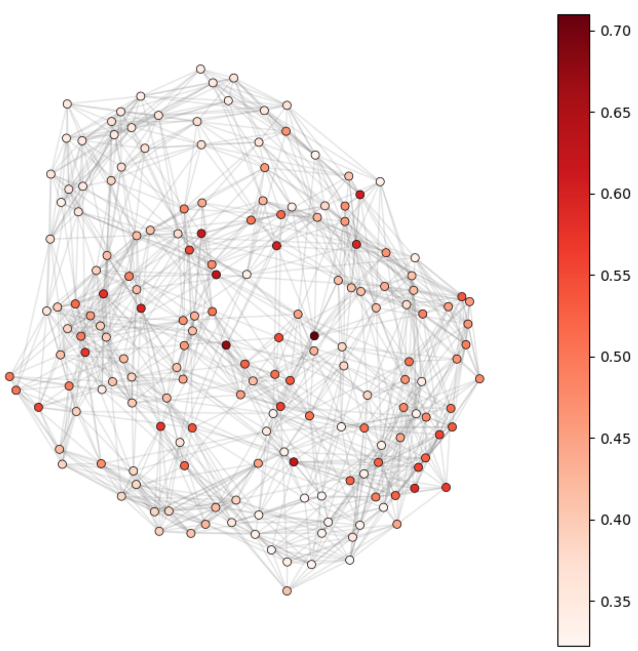


Figure A.3: Clustering performance on synthetic assortative graphs ($N = 250$, $\beta = 0.6$, $Q = 5$). **(left** Adjusted Rand Index (ARI) across noise levels ζ. **right** Partial Membership Error (PME) comparing soft assignments.

A.6 HCP-100 dataset



(a) Hard clusters on subject 0.



(b) Node-wise instability on subject 0.

Figure A.4: Cluster assignments and spatial instability in HCP-100 brain graphs.